

## SAND POINT, AK, USA

WMO: 703165

Lat: 55.319N

Lon: 160.521W

Elev: 21

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25617

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>12.0 | (c)<br>16.0 | (d)<br>1.0                    | (e)<br>5.8 | (f)<br>14.2 | (g)<br>4.7 | (h)<br>7.0 | (i)<br>16.0 | (j)<br>41.3           | (k)<br>31.1 | (l)<br>36.9 | (m)<br>24.6 | (n)<br>22.4           | (o)<br>340 | (p)<br>0.779 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>6.9             | (c)<br>62.7     | (d)<br>55.7 | (e)<br>60.6 | (f)<br>54.6 | (g)<br>57.3 | (h)<br>52.8 | (i)<br>57.3         | (j)<br>60.5 | (k)<br>55.8 | (l)<br>58.8 | (m)<br>54.5 | (n)<br>56.9 | (o)<br>8.7           | (p)<br>160 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 55.4                            | 65.3 | 57.7 | 54.5 | 63.2 | 57.0 | 53.8 | 61.5 | 56.4 | 24.6          | 59.7 | 23.7 | 58.6 | 23.0 | 57.0 | 65.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 34.2          | (b) 29.1 | (c) 25.8 | (e) 9.9                    | (f) 65.8 | (g) 3.9            | (h) 2.4 | (i) 7.0                                            | (j) 67.6 | (k) 4.8    | (l) 69.0 | (m) 2.6    | (n) 70.4 | (o) -0.3   | (p) 72.1 |    |    |
| (a) 34.2          | (b) 29.1 | (c) 25.8 | (e) 8.9                    | (f) 58.1 | (g) 3.9            | (h) 2.8 | (i) 6.1                                            | (j) 60.1 | (k) 3.8    | (l) 61.7 | (m) 1.6    | (n) 63.3 | (o) -1.2   | (p) 65.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 41.3   | 32.2  | 33.5  | 32.4  | 36.1  | 41.8  | 47.7  | 52.6  | 53.9  | 50.2  | 43.0  | 37.8  | 34.3  |  |  |
|                                                                        | DBStd       | 9.27   | 7.49  | 6.36  | 6.81  | 4.60  | 3.77  | 3.15  | 2.98  | 2.96  | 3.32  | 4.40  | 5.32  | 6.49  |  |  |
|                                                                        | HDD50       | 3436   | 553   | 463   | 545   | 416   | 255   | 83    | 3     | 36    | 219   | 367   | 488   |       |  |  |
|                                                                        | HDD65       | 8640   | 1018  | 883   | 1010  | 866   | 719   | 519   | 385   | 343   | 444   | 683   | 817   | 953   |  |  |
|                                                                        | CDD50       | 271    | 0     | 0     | 0     | 0     | 1     | 14    | 88    | 125   | 42    | 1     | 0     | 0     |  |  |
|                                                                        | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 10.9   | 11.9  | 11.7  | 11.9  | 11.2  | 9.4   | 8.0   | 7.6   | 8.7   | 10.9  | 12.9  | 13.9  | 13.1  |  |  |
|                                                                        | PrecAvg     | 41.4   | 3.9   | 3.5   | 2.8   | 2.3   | 2.6   | 3.0   | 3.7   | 3.0   | 4.4   | 3.8   | 4.0   | 4.4   |  |  |
| (15) Precipitation                                                     | PrecMax     | 56.3   | 6.3   | 7.3   | 5.8   | 3.7   | 4.9   | 4.9   | 5.7   | 4.4   | 7.2   | 6.5   | 6.6   | 9.4   |  |  |
|                                                                        | PrecMin     | 31.1   | 1.6   | 1.1   | 0.8   | 0.7   | 0.7   | 1.3   | 1.5   | 1.1   | 3.0   | 1.9   | 1.5   | 1.6   |  |  |
|                                                                        | PrecStd     | 6.1    | 1.3   | 1.5   | 1.3   | 0.8   | 1.0   | 0.9   | 1.1   | 0.8   | 1.1   | 1.2   | 1.3   | 1.7   |  |  |
|                                                                        | PrecStd     | 6.1    | 1.3   | 1.5   | 1.3   | 0.8   | 1.0   | 0.9   | 1.1   | 0.8   | 1.1   | 1.2   | 1.3   | 1.7   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 45.1  | 44.9  | 44.9  | 47.9  | 56.8  | 62.1  | 64.5  | 65.6  | 61.0  | 54.2  | 48.8  | 45.8  |  |  |
|                                                                        |             | MCWB   | 41.8  | 42.1  | 41.5  | 42.0  | 46.1  | 53.0  | 56.5  | 59.1  | 55.6  | 51.3  | 46.6  | 44.0  |  |  |
|                                                                        | 2%          | DB     | 43.1  | 43.1  | 42.9  | 45.1  | 52.4  | 57.2  | 62.6  | 63.0  | 57.3  | 52.2  | 47.2  | 45.0  |  |  |
|                                                                        |             | MCWB   | 40.4  | 40.4  | 39.6  | 40.1  | 44.4  | 50.4  | 55.4  | 56.5  | 53.6  | 49.8  | 44.9  | 42.6  |  |  |
|                                                                        | 5%          | DB     | 41.2  | 41.4  | 40.8  | 43.3  | 48.4  | 54.9  | 59.4  | 60.8  | 55.5  | 50.3  | 45.9  | 43.3  |  |  |
|                                                                        |             | MCWB   | 38.7  | 39.0  | 37.9  | 39.2  | 43.1  | 49.4  | 53.4  | 55.4  | 51.8  | 47.4  | 43.1  | 40.4  |  |  |
|                                                                        | 10%         | DB     | 39.4  | 39.8  | 39.2  | 42.5  | 46.3  | 53.6  | 57.1  | 57.5  | 54.7  | 48.4  | 45.1  | 42.5  |  |  |
|                                                                        |             | MCWB   | 37.0  | 37.4  | 36.4  | 38.8  | 41.8  | 48.6  | 52.3  | 53.5  | 51.3  | 45.6  | 42.1  | 39.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 43.7  | 43.3  | 42.7  | 43.6  | 48.2  | 54.8  | 58.9  | 60.1  | 57.5  | 52.5  | 47.4  | 45.5  |  |  |
|                                                                        |             | MCDB   | 44.4  | 44.1  | 43.9  | 45.9  | 55.3  | 59.7  | 62.5  | 63.8  | 58.6  | 52.9  | 47.9  | 45.5  |  |  |
|                                                                        | 2%          | WB     | 41.1  | 41.3  | 40.3  | 41.4  | 46.0  | 52.1  | 56.7  | 57.8  | 55.3  | 50.3  | 45.8  | 43.6  |  |  |
|                                                                        |             | MCDB   | 42.9  | 42.7  | 42.4  | 44.0  | 49.9  | 55.8  | 60.6  | 61.2  | 56.7  | 51.4  | 46.4  | 44.6  |  |  |
|                                                                        | 5%          | WB     | 39.3  | 39.7  | 38.5  | 40.3  | 44.5  | 50.5  | 55.0  | 56.4  | 53.8  | 47.8  | 44.4  | 41.4  |  |  |
|                                                                        |             | MCDB   | 41.2  | 41.6  | 40.4  | 43.1  | 47.7  | 54.3  | 58.2  | 59.4  | 55.0  | 49.1  | 45.7  | 43.7  |  |  |
|                                                                        | 10%         | WB     | 38.1  | 38.3  | 37.5  | 39.0  | 43.0  | 48.9  | 53.7  | 55.1  | 52.4  | 46.3  | 42.3  | 39.8  |  |  |
|                                                                        |             | MCDB   | 39.9  | 40.0  | 39.1  | 41.8  | 46.3  | 52.5  | 56.2  | 57.8  | 54.3  | 47.8  | 44.9  | 42.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.0   | 6.1   | 6.8   | 7.0   | 7.6   | 7.4   | 7.2   | 6.9   | 6.5   | 6.0   | 6.0   | 5.9   |  |  |
|                                                                        |             | MCDBR  | 5.9   | 5.9   | 7.0   | 7.9   | 11.9  | 12.1  | 11.3  | 10.1  | 7.5   | 6.6   | 6.7   | 6.1   |  |  |
|                                                                        |             | MCWBR  | 5.5   | 5.7   | 5.6   | 5.8   | 7.2   | 7.2   | 6.5   | 6.0   | 5.3   | 5.5   | 6.7   | 6.2   |  |  |
|                                                                        | 5% WB       | MCDBR  | 5.6   | 5.9   | 6.3   | 7.3   | 9.7   | 10.7  | 9.3   | 8.2   | 6.4   | 6.0   | 6.1   | 6.0   |  |  |
|                                                                        |             | MCWBR  | 5.4   | 6.0   | 5.8   | 5.6   | 6.4   | 6.7   | 5.9   | 5.6   | 5.8   | 5.7   | 6.8   | 6.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.282 | 0.330 | 0.353 | 0.385 | 0.390 | 0.391 | 0.374 | 0.383 | 0.366 | 0.346 | 0.328 | 0.287 |  |  |
|                                                                        | taud        |        | 2.399 | 2.302 | 2.249 | 2.221 | 2.280 | 2.357 | 2.473 | 2.428 | 2.404 | 2.317 | 2.177 | 2.336 |  |  |
|                                                                        | Ebn at Noon |        | 213   | 234   | 256   | 264   | 270   | 271   | 274   | 264   | 252   | 223   | 175   | 178   |  |  |
|                                                                        | Edh at Noon |        | 16    | 25    | 33    | 39    | 39    | 36    | 32    | 31    | 28    | 24    | 19    | 13    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 162   | 329   | 753   | 1146  | 1409  | 1515  | 1352  | 1122  | 811   | 460   | 206   | 113   |  |  |
|                                                                        | RadStd      |        | 21    | 38    | 117   | 69    | 170   | 173   | 146   | 105   | 70    | 35    | 20    | 12    |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (5 neighbors) | N/A   | +2.61   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page